

The Science-based Playbook of **Pricing & Promotions**

Ariyh“

Boost your profits by putting into action the latest scientific discoveries from the world's best business schools



Created by Thomas McKinlay,
Founder of Ariyh

Sold to
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About this playbook

This playbook is a practical guide written to be as short, clear, and effective as possible. Not a long-winded, time-consuming read full of filler information.

The focus is on the most useful, latest, scientific discoveries. It is not meant to cover principles of pricing and promotions that have been known for decades (e.g. anchoring, decoy effect).

Most of the techniques covered here have not yet been 'translated' from research into practice by mainstream books, trainings, or consultants. It's highly unlikely that your competitors are aware of them.

To make the most of this playbook, return to it regularly to make sure you are optimizing your profits, or even to price a new product from scratch.

All scientific research covered here has survived rigorous statistical analyses, peer-review (i.e. been picked apart and challenged by other researchers), and been published in a reputable scientific journal. Most research papers are never accepted for publication.

One last thing. For weekly 3-min practical summaries of the latest scientific discoveries, you can subscribe to Ariyh at no cost. Just drop your email on [ariyh.com](mailto:brian@briancliette.me), if you haven't already.

Some of the institutions we should be grateful for the research covered in this playbook:





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NOTE

The next sections are techniques to optimize and tweak your existing price to make it more appealing.

But if you are not confident that you have chosen the right price in the first place you are leaving money on the table. To find the price range that is most profitable for your product (e.g. \$15 vs \$25) start from the last section of the playbook (#5).

“ 1.

Optimize the digits of your price

Should your price be rounded (\$220), just-below (\$199), or precise (\$223.52)?

Rounded pricing

Some form of rounded pricing (e.g. \$80, \$6) is usually your best bet.



What it is

- Rounded prices end with 0s or feel whole and simple. For example: \$160 or \$20.
- For our purposes here we will also consider numbers like \$35, \$1, or \$18 as rounded.



How it works

- People think and communicate in rounded prices, so they're much easier to mentally process.
- This makes decisions effortless and easier to take, increasing purchases.



What it does

Good:

- People like rounded prices more.
- Associated with quality.
- Easy to mentally process.

Bad:

- May not maximize your profit.

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Use it when

- Quality is the focus in the buying decision, rather than price. For instance:
 - Premium or luxury products
 - Purchases involve risk (e.g. medical, safety-related)
 - Most services (e.g. accounting services, plumbing)
- You want to cultivate a premium brand image
- The purchase is mainly feelings or pleasure-driven (hedonic products, e.g. spa, flowers)
- You want to make choices easier and more convenient for your customers (e.g. to improve your UX)
- Your product is negotiable and you expect people to bid above your asking price (e.g. a tight real estate market).
- When in doubt about your ideal pricing.

Just-below pricing

The latest evidence shows that just below prices (e.g. \$9.99) aren't always the right choice.



What it is

- Just-below prices use amounts that are below, but close to, a rounded number.
- For example, \$2.99 vs \$3.00, \$297 vs \$300, or \$4,950 vs \$5,000



How it works

- Because of the [left-digit bias effect](#), people focus more on the first digit than the last ones. The left-digit of the amount [should go down](#) for it to be effective (the difference between
- \$1.99 vs \$3.00 feels much bigger than \$2.00 vs \$2.99, although it's only \$0.02 different). The price discount seems [larger when the left digit is small](#) (e.g. 1 or 4 vs 7).



What it does

Good:

- Signals good value.

Bad:

- Associated with lower quality.
- Harder to mentally process and remember.



Use it when

- Running a price promotion (e.g. was \$55, now \$39.99)
- [Price is the focus in the buying decision](#), rather than quality
- You want to cultivate a low price brand image
- Your price is low and the purchase is low risk (e.g. groceries)



Special cases

- If prices are 4 figures or more, common for many non-US Dollar \$ or Euro € currencies (e.g. Indian Rupee ₹, Japanese Yen ¥, Turkish Lira ₺), [repeating the last 3 digits overrides the just-below effect](#) (e.g. \$3,222). Instead, repeat the last 3 digits with a number below 5 (ideally 1) to give a feeling of better value. For example, use ¥2,111 instead of ¥1,999.
- While the 9 ending is popular in western countries, 8 endings are more popular [in Japan](#) (where 8 is a lucky number) and [in China](#) (where 8 represents prosperity).
- Don't use it when you want customers to [upgrade to a more expensive option](#) (try to keep the two prices on the same left-digit level, e.g. \$40.50 vs \$48.50 for the upgrade)

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Precise pricing

Precise prices (e.g. \$2,347.52) can sometimes send useful signals.



What it is

- Precise prices use non-rounded amounts that seem deliberately calculated, or in some contexts can seem random.
- They can be moderately precise (e.g. \$981,200) or highly precise (e.g. \$981,218.37).
- They're different from just-below prices (e.g. \$2.72 or \$3.26 are precise, \$2.99 or \$2.95 are just-below).



How it works

- Precise prices give the impression of being calculated that way for a reason, likely by an expert, so they feel more legitimate.
- In negotiations, they are strong anchors because they nudge people to think and negotiate in smaller chunks (e.g. by \$2.50 if the price is \$52.50 vs by \$10 if it's \$50).



What it does

Good:

- Signals [competence](#) and [confidence](#) of the seller.
- Feels more rational.

Bad:

- Very hard to mentally process and remember.
- Can cause mental overload and turn people away.



Use it when

- It's important to signal rationality, expertise, and confidence in big decision purchases (e.g. ad hoc B2B contract, flat-rate home refurbishments).
- You expect people to try to negotiate down your opening price (e.g. used cars, real estate, B2B contracts). [You might get fewer offers](#), but they'll be higher.
- You want an alternative way from just-below prices to set prices that are slightly higher than rounded ones (e.g. \$3.76 vs \$3.50). Be aware that the evidence that this works is not as strong as the evidence for just-below prices.



Special cases

- Very precise prices (e.g. \$23,721.85) can [backfire in negotiations with expert buyers](#) (e.g. a purchasing manager). When selling to an expert, justify the precise price (e.g. it was evaluated by a surveyor) or stick to moderately precise prices (e.g. \$23,750).

⚡ Examples of applications

Product	Context	Recommended technique	Why
Freemium language learning app (e.g. Duolingo)	Trying to encourage free version users to upgrade to the premium version	Just-below (e.g. \$4.99 / month)	Users have already experienced the app's quality so signaling value is more important to encourage the leap
Flatscreen TV	In a retailer with a brand focused on value (e.g. Walmart)	Low last 3-digits (e.g. \$1,222)	For prices of 4 figures or more, repeating a low number in the last 3 digits shows more value than just-below pricing
Premium brand shoes (e.g. Nike)	Special Black Friday discounted price	Just-below (e.g. \$79.99)	The price promotion overrides the need to signal quality and premiumness through a rounded price
Enterprise accounting software	Negotiating an ad hoc package for a large buyer	Precise (e.g. \$53,832 / month)	Signals expertise and makes counter offers higher. Must be justified (e.g. with a breakdown of costs for each service)
Online course to master LinkedIn	From an individual sharing their expertise (e.g. not a well-known university)	Rounded (e.g. \$150)	Use every opportunity to signal quality, this is a key way to do that
Antique motorbike for auction	Whoever bids higher buys it. High demand expected.	Rounded (e.g. \$15,000 opening price)	Signals the premiumness of the bike. Encourages people to bid in multiples of 1,000s
Run-down used motorbike for sale	Trying to get offers as close to asking price as possible, but they will probably be lower	Precise (e.g. \$2,865 asking price)	Signals that the price is right. People will bid lower in smaller increments (e.g. \$265 lower)

“ 2.

Further optimize your price

The latest techniques to make your full-price more appealing.

Use flat-rates when possible

Extended Ariyh Insight



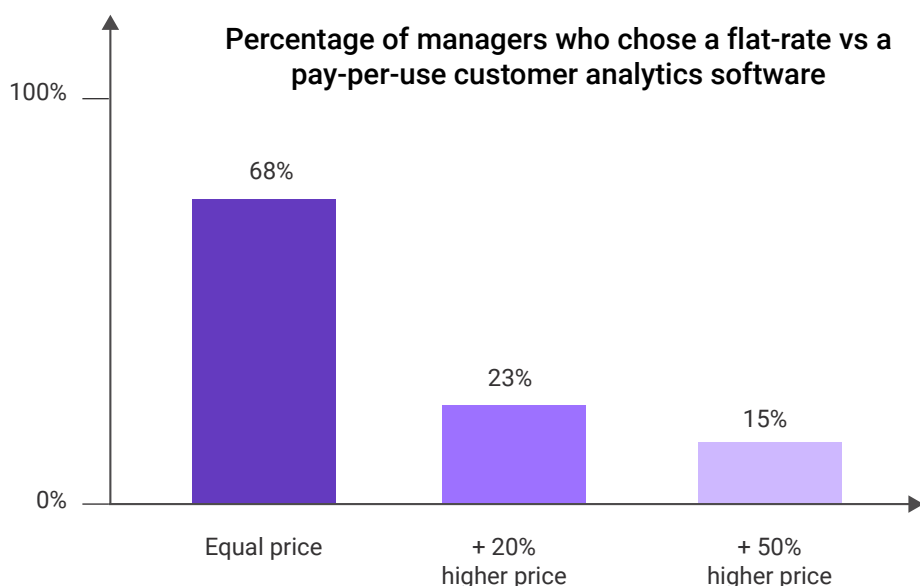
Recommendation

- Offer flat-rate prices that include most or all the options customers may want (e.g. \$50/month for 100GB of mobile data plans, unlimited revisions by a designer).
- Charge more for your flat-rate plans compared to pay-per-use (e.g. hourly rates), or expect higher demand if you don't. You can keep pay-per-use for entry-level customers that want a low-commitment trial run.
- For example, a fixed price of \$6,000 is often more attractive than paying \$50 an hour for an expected 100 hours (= \$5,000).



Why?

- People overestimate their usage and want to ensure they won't pay more than planned.
- They also hate seeing costs gradually rise (i.e. the taximeter effect).
- Flat-rates are more convenient and simple to manage, and reduce the feeling of possible fraud (e.g. a taxi taking a longer route on purpose).



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CASE STUDY

Marketing agency [Outlier Creative](#) used this insight when they received a request for proposal for an event by the European Commission.

Instead of offering per hour rates for individual services (e.g. \$65/hour for website copywriting), they simplified and let the client choose between 3 different flat-rate packages (e.g. a \$4,490 basic starter pack or \$14,490 if they wanted advanced features such as 3D video).

They believe this approach was key in landing one of their biggest clients.



Research

- [Purchasing professionals and the flat-rate bias: Effects of price premiums, past usage, and relational ties on price plan choice](#). Journal of Business Research (April 2021).
- Researchers
 - Mario Kienzler. Linköping University. Sweden
 - Christian Kowalkowski. Linköping University and Hanken School of Economics. Sweden and Finland
 - Daniel Kindström. Linköping University. Sweden

Keep price components simple

Extended Ariyh Insight



Recommendation

- When you have to price components separately (e.g. for additional km of a rental car, additional users of a SaaS), keep prices simpler than those of your competitors.
- Use rounded prices (e.g. \$5, not \$4.61), similar price endings for different components, and make options easily comparable. People will be willing to pay more.
- For example, as a telecom provider, charge \$0.10 per minute and \$0.10 per extra GB of data, instead of \$0.07 per minute, \$0.09 per extra GB, and a \$6/month service charge.



Why?

- When prices are complex, people usually decide it's not worth the mental effort to calculate the actual price, and just assume you are more expensive than simpler options.
- If they do decide to calculate it, they find it unfair that you made them put in such effort. It's even worse if they suspect you made it difficult on purpose.



Research

- How price complexity takes its toll: The neglected role of a simplicity bias and fairness in price evaluations. Journal of Business Research (June 2013).
- Researchers
 - Christian Homburg. University of Melbourne and University of Mannheim.
 - Australia and Germany.
Dirk Totzek. University of Passau. Germany
 - Melanie Krämer. University of Mannheim. Germany

Show prices in red to men

Extended Ariyh Insight



Recommendation

- If at least some of your customers are men, write your prices in red to make them seem better value. For example, in one experiment, men evaluated red prices as 66% cheaper.

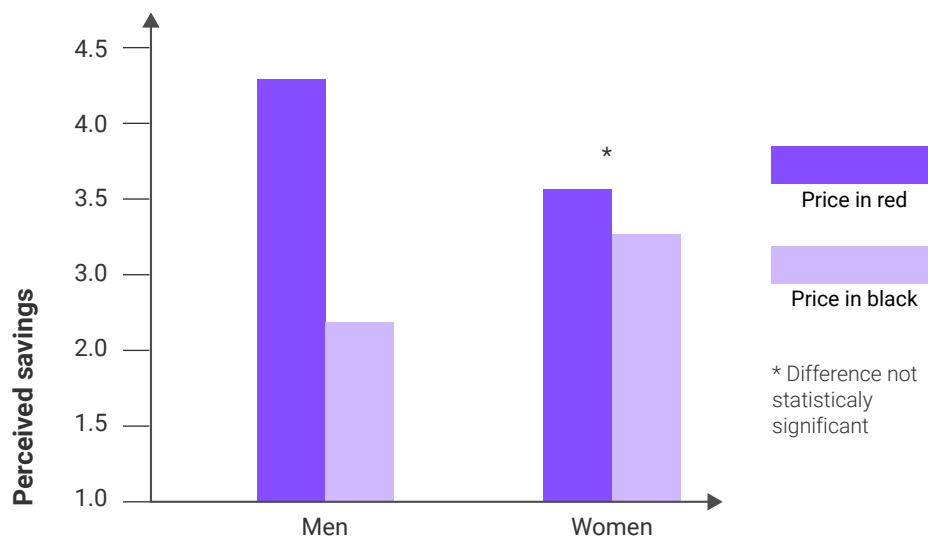


Why?

- Men, in particular, are attracted to red because it's associated with testosterone and dominance, and attracts attention. For example, [referees are biased towards teams that wear red](#), and men are [more interested in asking a woman out](#) if she's wearing red.
- Women are unaffected, one reason being that they evaluate ads more in-depth than men.

! Careful

- All prices must be in red. If only one or a few prices are red [the effect might backfire](#).



Research

- [Are men seduced by red? The effect of red versus black prices on price perceptions.](#) Journal of Retailing (February 2013).
- Researchers:
 - Nancy Puccinelli. Saïd Business School, Oxford University. UK
 - Rajesh Chandrashekar. Silberman College of Business, Fairleigh Dickinson University. USA
 - Dhruv Grewal. Babson College. USA
 - Rajneesh Suri. Le Bow College of Business, Drexel University. USA

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⚡ Examples of applications

Product	Context	Recommended technique	Why
Discount grocery store	A store promising the lowest prices	Prices in red	Even if only part of all customers are men, they will perceive the store as better value
Wedding venue	Normally offering several commonly chosen extras (e.g. food, flowers, cake)	Flat-rate (e.g. €150 per guest, all included)	Becomes more attractive than other cheaper options with complex choices
Website design and hosting	Customers choose which services they want included. There is too much variance to offer flat-rates	Simple price components (e.g. \$10 / month extra 10,000 visits, \$10 per extra 10GB)	Simple prices make the difficult choices of which features and quantities to choose easier.

“ 3. Price promotion techniques

The latest techniques to make your price promotions more appealing.

“10% off” vs “\$10 off”

Extended Ariyh Insight



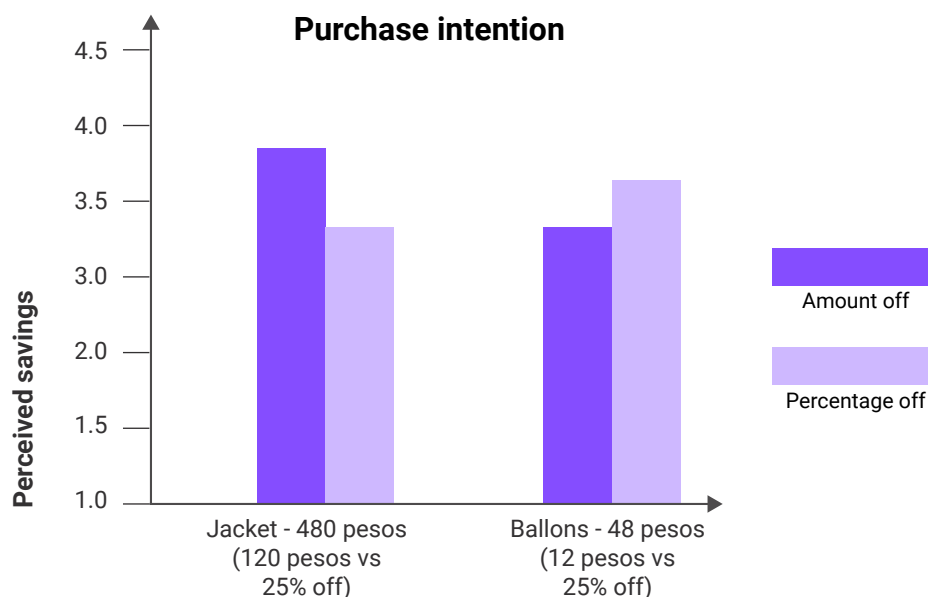
Recommendation

- When discounting a full price that's higher than 100 (in any currency), use an amount off (e.g. \$30 off).
- For prices below 100, use a percentage off (e.g. 40% off).



Why?

- Mathematically, the number used in the discount looks bigger.
- Bigger discount numbers instinctively seem better.
- For example, 20% off \$400 is \$80, and 80 is a much bigger number than 20.



Research

- [Amount off versus percentage off—when does it matter?](#). Journal of Business Research (August 2015).
- Researchers:
 - Eva González. EGADE Business School. Mexico
 - Eduardo Esteva. EGADE Business School. Mexico
 - Anne Roggeveen. Babson College. USA
 - Dhruv Grewal. Babson College. USA

“Was 25% higher” vs “Now 20% lower”

Extended Ariyh Insight



Recommendation

- Use the already discounted price, not the original price to calculate the discount amount.
- For example, if discounting by \$2 a full price of \$10, you would normally say ‘now 20% lower’ ($\$2/\$10 = 20\%$ lower). Instead, use the already discounted price of \$8 ($\$10 - \2 discount) and say ‘was 25% higher’ ($\$2/\$8 = 25\%$ higher). It’s now a larger %.
- This can also be used to announce price increases (e.g. the price will increase 50%) or product features (e.g. 40% longer battery life).



Why?

- The number used in the % change looks bigger.
- Bigger discount numbers instinctively seem better.



Special cases

- The effect weakens or disappears when percentages go over 100% (e.g. was 125% higher) because [people ignore the ‘100%’ and only count the ‘25%’](#). For example, a \$12 discount on an item that was \$20 ($\$12/\$8 = 150\%$ difference) is perceived as 50% higher, not 150% higher as it actually is.



Research

- [Reframing the discount as a comparison against the sale price: does it make the discount more attractive?](#). Journal of Marketing Research (June 2018).
- Researchers:
 - Abhijit Guha. Darla Moore School of Business, University of South Carolina. USA
 - Abhijit Biswas. Wayne State University. USA
 - Dhruv Grewal. Babson College. USA
 - Swati Verma. Lawrence Technological University. USA
 - Somak Banerjee. Wayne State University. USA
 - Jens Nordfält. Stockholm School of Economics. Sweden

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Store-wide vs product-specific discounts

Extended Ariyh Insight



Recommendation

- When running a store-wide promotion for all or most products, use a percentage % off discount (e.g. 'Today everything 15% off'). Sales increased 42.3% in one experiment.
- If the discount is product or brand-specific, use \$ amount off (e.g. \$2 off Gillette razors).



Why?

- Percentage % off discounts remove limits of taking advantage of the discount, so people enter a "the more I spend, the more I save" mindset and buy more.
- Amount \$ off discounts clearly show how much people will miss out on if they don't use it (e.g. \$10). They also don't need to calculate the savings by themselves (unlike with % discounts).



Careful

- Some customers may use the discounts to hoard their purchases and wait longer until the next time they buy. If you can, try to send these offers only to segments that will buy new products (e.g. items they wouldn't usually buy), not just hoard.



Research

- [Empirical decomposition of customer responses to discount coupons in online FMCG retailing.](#) Journal of Retailing and Consumer Services (October 2020).
- Researchers:
 - Samaneh Montazeri. Deakin Business School, Deakin University. Australia
 - Ali Tamaddoni. Deakin Business School, Deakin University. Australia
 - Stanislav Stakhovych. Monash Business School. Australia
 - Michael Ewing. Deakin University. Australia

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Keep the same price ending for discounts

Extended Ariyh Insight



Recommendation

- When offering a discounted price (e.g. \$16.75), keep the price ending consistent with the original price (e.g. \$26.75, not \$26.99).
- Always show the original price next to the new one so people understand how much they are 'gaining' with the offer.



Why?

Subtractions are easier (e.g. discount is \$20 = \$36.95 - \$16.95), so people are more likely to analyze and consider the benefits of the promotion, and are more likely to buy.



Research

- [Consistent price endings increase consumers' perceptions of cheapness](#). Journal of Retailing and Consumer Services (May 2021).
- Researchers:
 - Hui-Hsi Hung. Minjiang University. China
 - Yin-Hui Cheng. National Taichung University of Education. Taiwan
 - Shih-Chieh Chuang. National Chung Cheng University. Taiwan
 - Annie Pei-I Yu. National Chung Cheng University. Taiwan
 - Yu-Ting Lin. National Chung Cheng University. Taiwan

Big discounts backfire

Extended Ariyh Insight



Recommendation

- Don't discount prices more than 40%, unless there is an exceptional reason (e.g. store shut down clearance).
- Discounts of 60% or more tend to reduce rather than increase sales, especially if your brand is not well known or it's hard to judge a product's quality before buying it.



Why?

- Big discounts signal low product quality, which puts off most people from buying.
- The effect is so strong that it can trigger a [negative placebo effect](#). For example, people who drank an energy drink sold at a discounted price solved fewer puzzles.



Research

- [When discounts hurt sales: The case of daily-deal markets](#). Information Systems Research (July 2018).
- Researchers
 - Zike Cao. Rotterdam School of Management, Erasmus University. Netherlands
 - Kai-Lung Hui. School of Business and Management, Hong Kong University of Science and Technology. Hong Kong
 - Hong Xu. School of Business and Management, Hong Kong University of Science and Technology. Hong Kong

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⚡ Examples of applications

Product	Context	Recommended technique	Why
Fashion ecommerce store	Trying to encourage as many sales as possible for Black Friday	Store-wide % coupon (e.g. 25% off all products)	Encourages people to buy more items at once
Home security system	Company trying to sell off stock of old models	Discount at maximum 40%, not more	A higher discount would hurt credibility in the quality of the product too much
Chocolate bar	Running an in-store promotion in a department store	Product-specific \$ coupon (e.g. \$2 off) or	Makes it clear how much the buyer gains from the offer
Website analytics software	Offering a special promotion to upgrade to premium plans	Keep the same price endings (e.g. was \$9.99, now \$6.99 / month)	Easier to calculate how much the buyer gains from the offer
Jeans	A \$60 pair of jeans are on offer for \$15 less	Percentage off discount (25% off)	The discount appears bigger, making people more likely to buy
Portable speaker	Special \$30 discount for Cyber Monday on a full price of \$90	Was % higher (e.g. was 50% higher, not now 33% lower)	The discount appears bigger, making people more likely to buy

“ 4.

Beyond price

The latest techniques to make your promotions more attractive, without changing the price

Boost sales with pick-your-price

Extended Ariyh Insight



Recommendation

- Let customers choose from 3 or 4 price options how much they want to pay for your product (e.g. \$25, \$35, \$50, or \$100).
- People will be more likely to buy. Keep in mind they will usually choose the lowest price. Don't use it if people are buying multiple products at once (e.g. fashion, groceries).



Why?

- Pick-your-price gives a feeling of personal control. People like that, so they are more likely to buy.



Careful

- Pick-your-price is different from pay-what-you-want, which lets people choose any amount they like (e.g. choose to pay any price, including zero).
- Do not use pay-what-you-want. Research has repeatedly found that it does not work. It's so mentally demanding that it actually reduces sales. The only situation it might come in handy is when asking for donations.



Research

- The Control–Effort Trade-Off in Participative Pricing: How Easing Pricing Decisions Maximizes Pricing Performance. Journal of Marketing (January 2021).
- Researchers
 - Cindy Xin Wang. Orfalea College of Business, California Polytechnic State University. USA
 - Joshua Beck. Lundquist College of Business, University of Oregon. USA
 - Hong Yuan. Lundquist College of Business, University of Oregon. USA

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Use “For \$0”, not “For free”

Extended Ariyh Insight



Recommendation

- When giving something for free, don't say it's “Free”.
- Instead, say it's “For \$0”, “At no cost”, or similar. Your conversions will be higher.



Why?

- “For \$0” reminds people that they are saving money, which makes them value the product more.
- It also activates people's loss aversion (i.e. we feel more strongly and pay more to avoid losing something rather than gaining the same thing). “Free” suggests a gain, while “For \$0” reminds people of the money they are not losing - which is more powerful.



Research

- [Is \\$0 Better than Free? Consumer Response to “\\$0” versus “Free” Framing of a Free Promotion.](#)
Journal of Retailing (December 2019).
- Researchers
 - Jieun Koo. Pukyong National University. South Korea
 - Kwanho Suk. School of Business, Korea University. South Korea

Encourage pre-orders with gifts

Extended Ariyh Insight



Recommendation

- To encourage pre-orders for far away launches (e.g. 9 weeks) give a gift to customers who pre-order (e.g. a complementary product for free). It is much more effective than giving price discounts.
- For pre-orders of products that are about to launch (e.g. in 1 week), both gifts and discounts are effective.
- Keep a short pre-launch window for products that appeal to rationality (e.g. utilitarian products like house appliances). Emotion-driven products are less sensitive to long pre-launch time windows (e.g. experiences, fashion).



Why?

- Free gifts encourage a positive feeling of anticipation for a product's release.
- Discounts are rational, boring, monetary rewards. People prefer immediate rewards rather than future ones, so investing in future savings is a less interesting benefit.



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CASE STUDY

If you purchased this playbook during the pre-order window of 5-weeks before launch, you received an exclusive Ariyh insight as a gift (i.e. a secret newsletter that will never be sent).

It's impossible to reliably tell whether this technique played a factor or not. However, pre-orders were 56% higher than projected, and the exclusive gift may have been an additional motivator.



Research

- [The effect of positive anticipatory utility on product pre-order evaluations and choices](#). Journal of the Academy of Marketing Science (September 2021).
- Researchers
 - Amaradri Mukherjee. The School of Business, Portland State University. USA
 - Ronn Smith. College of Business, University of Wyoming. USA
 - Scot Burton. Walton College of Business, University of Arkansas. USA

Make the main product the 'gift' in a bundle

Extended Ariyh Insight



Recommendation

- Create a bundle with your product and a complementary product (e.g. a fitness tracker and a weighing scale). Then, frame the main product (e.g. the fitness tracker) as the gift (e.g. Buy the weighing scale and get the fitness tracker free).
- Try to target the offer to people that are looking for the product that's framed as a gift (e.g. if someone searches for "fitness tracker" they get shown the weighing scale + tracker offer).
- Note that this wasn't tested (and probably does not work) with products of very different value (e.g. buy an iPhone case and get an iPhone).



Why?

- People don't expect to find the product they're looking for as a free gift.
- When they come across the offer, they feel lucky and are more likely to buy it.



Research

- [Feeling Lucky: How Framing the Target Product as a Free Gift Enhances Purchase Intention](#). International Journal of Research in Marketing (October 2021).
- Researchers
 - Maggie Wenjing Liu. Tsinghua University. China
 - Chuang Wei. Tsinghua University. China
 - Lu Yang. Nanjing Agricultural University. China
 - Hean Tat Keh. Monash Business School, Monash University. Australia

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Use time limits to encourage action

Extended Ariyh Insight



Recommendation

- To encourage customers to buy now rather than wait (and probably forget), give a time limit for how long the product will be available or on offer.
- For example, limit the offer to 48 hours, or until a specific time and date (make sure to specify the timezone).



Why?

- This pushes people to take action at the moment that they're already most likely to buy.
- Reminding people they could lose the offer if they wait increases the intensity of emotions (e.g. feelings of hope, higher testosterone levels), which increases interest.

! Careful

- Companies often limit the number of products available (e.g. only 1,000 pieces), rather than the time they're available for.
- This method has been found to have a strong negative impact on people who miss out on the offer. They get angry with you and are likely to switch to a competitor. This is not the case if the offer was time-limited.



Research

- [The downside of scarcity: Scarcity appeals can trigger consumer anger and brand switching intentions](#). Psychology & Marketing (April 2021).
- Researchers
 - Alessandro Biraglia. Leeds University Business School. UK
 - Bryan Usrey. Norwich Business School, University of East Anglia. UK
 - Aulona Ulqinaku. Leeds University Business School. UK

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Keep showing some sold-out options

Extended Ariyh Insight



Recommendation

- If you sell different products or your product has various configuration options and some of them sell out, keep showing them - marked as sold-out.
- When you have too many options, the effect backfires, so it's better to remove some sold-out options from sale or stock up.
- Try to keep the proportion of sold-out options between 10% and 30%.



Why?

- [Phantom options](#) (options still displayed but that we can't choose) give powerful signals.
- People tend to assume that others have better information than they do. So if others bought them to the point of selling out, the product must be good.
- But people react negatively when they feel they don't have freedom of choice. They are then likely to abandon or defer their purchase.



Research

- [A good way to boost sales? Effects of the proportion of sold-out options on purchase behavior.](#) International Journal of Research in Marketing (April 2021).
- Researchers
 - Jing Tian. Agricultural Bank of China. China
 - Rong Chen. Tsinghua University. China
 - Xiaobing Xu. Hainan University. China

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⚡ Examples of applications

Product	Context	Recommended technique	Why
Videogame	Available for pre-order 2 months before launch	Give gifts to who pre-orders (e.g. additional content or customizations)	Drives excitement about the game and grows pre-orders more than a discount
Smartphone	Sold on a website where people can choose from several options (e.g. colors, memory sizes)	Keep showing some less popular options as sold out when you run out of stock of them	Showing less popular options as sold out encourages more sales of the ones that are available (and people wanted anyway)
Gym membership	Special offer to encourage a wave of signups	Use a time-limited price promotion (e.g. join before 25th April for 30% off)	People will be more motivated to take action when coming across the offer and won't blame the gym if they miss it
Technical SEO guide	SEO tool company offering gated content to collect leads	Get the guide for \$0	Reminds people that the guide has value, increasing conversions
Hot dogs	A hot dog cart trying to encourage more sales of its main product	Buy a large drink, get a hot dog free	Makes people feel lucky and more likely to buy the bundle
Sunglasses	A hyped-up limited edition model in collaboration with an upcoming artist	Pick-your-price	Drive further hype, reach, and sales

“ 5.

Guide: How to find your optimal price

This section is a step-by-step guide to research the best price for your product. It uses a new scientific method to understand how much people are willing to pay, without having to hire an expensive market research company.

It requires specific steps to be executed correctly and accurately. So here is an in-depth guide of how to apply it to your own product.

The data we need

We want to choose the price for your product that generates the highest profit. This is the point where Number of sales x Price (after costs) is highest.

The optimal number (e.g. \$60 profit per day) can be reached in different ways. For example:

- **LOW PRICE + HIGH VOLUME:** 100 people per day would buy a cookie if it's priced at \$0.80 (cost: \$0.20 per cookie) Profit: \$60 per day (\$0.60 x 100)
- **HIGH PRICE + LOW VOLUME:** 20 people per day would buy a cookie if it's priced at \$3.20 (cost: \$0.20 per cookie). Profit: \$60 per day (\$3.00 x 20)
- **MID PRICE + MID VOLUME:** 50 people per day would buy a cookie if it's priced at \$1.40 (cost: \$0.20 per cookie). Profit: \$60 per day (\$1.20 x 50)

To figure out which options you have from the above, we need to find out a key variable:

👉 **How many people would buy your product at each different price?**

The latest scientific method to find it

Scientists [discovered in 2020](#) a new method that is easy enough for anyone to apply while accurate enough for the vast majority of situations. We will apply it step-by-step to find your most profitable price.

Other methods that have been around for a while tend to be either very difficult and expensive to execute, or very inaccurate (and often both). For example:

- The [Becker–DeGroot–Marschak method](#) from 1964 is the most accurate technique. But it involves selling your actual product and is so complex and expensive that very few companies use it.
- A simple “How much would you pay?” question is easy to ask and the most commonly used by market research firms. It’s better than nothing, but answers are notoriously inaccurate by a large margin (e.g. 50%).
- Many companies simply resort to looking at what prices competitors or similar products have, but this is similar to just flipping a coin. It’s far from profit maximization.

How it works

- This method found a simple statistical trick to make the “How much would you pay?” approach much more accurate.
- When people answer “How much would you pay at a maximum?” they tend to overestimate how much they would truly pay.
- When people answer “Would you buy the product for \$X?” their answers are biased by how low or high the price is because they see it as an indicator of quality.
- Combining the answers from the two questions above gives an accurate, unbiased result of how much people would pay.

Research

- [A de-biased direct question approach to measuring consumers' willingness to pay](#). International Journal of Research in Marketing (June 2020).
- Researchers
 - Reto Hofstetter. University of Lucerne. Switzerland
 - Klaus Miller. Goethe University Frankfurt. Germany
 - Harley Krohmer. University of Berne. Switzerland
 - John Zhang. The Wharton School, University of Pennsylvania. USA

Steps to take

These are the steps you should follow to apply the method. We will go into how to execute each one in practice in the case study below.

1. Create two surveys with one question each:

Survey A - Open question survey that asks: "How much would you pay at a maximum for [product]?"

Survey B - Closed question survey that asks: "Would you buy [product] for \$X? [Yes/No]"

- This question will randomly use a price from a set of options that we will feed it (e.g. "Would you buy [product] for \$10 / \$15 / \$20?")
- Prices should have equal increments between them (e.g. \$10 increments, so \$20, \$30, \$40. Not \$20, \$50, \$100)
- Start with the lowest price you can imagine for your product and end with the highest. Be generous
- Use 7 different price options. You can use more, but if you do you will need more respondents (minimum 30 for each price)

2. Describe your product at the beginning of each survey as realistically as possible

(e.g. an extensive description, images, a sample, a video)

3. Recruit a large enough sample of your target customers.

- For best results, you want around 500 responses if you're using 7 price options (30 for each closed question = 210 + the same total size for the open question = 210 + safety margin for some that won't complete the survey properly or at all = 80)
- Use existing or prospective customers whenever possible (e.g. put the survey on your website, in your newsletter). If you can't, recruit volunteers from surveying platforms (e.g. on [Amazon MTurk](#)), although this is not ideal
- Give meaningful rewards to encourage answers (e.g. a coupon, a sweepstake for Amazon gift cards).
- Make sure to mention it's just 1 quick question to increase response rates

4. Run the surveys and collect results until you hit the desired number of answers

5. Combine the results of both surveys to understand how much each person in the open-ended question would truly pay

(see the case study below for how exactly to calculate these):

- Take the price they said they would pay
- Subtract the average price from all open questions
- Add the average price from all closed questions

6. Calculate what % of people would pay up to a certain price, and how many wouldn't

(e.g. 55% of people would pay \$100 or less, 22% would pay \$150 or less)

7. Pick your best price based on your strategy

(e.g. high prices + low sales volume)

Case study: Researching the best price for Ariyh's playbook

How I determined the best possible starting prices for the Playbook you are reading right now

Setting up and running the surveys

Requirements

- We need to randomly show Survey A ("How much would you pay at a maximum?") to 50% of participants and Survey B ("Would you buy [product] for \$X? [Yes/No]") to the other 50%.
- The 50% who answer Survey B, need to be randomly shown one of the 7 (or more) prices that we have chosen
- So we need a way to first randomly split participants 50:50 (Survey A and B), and then split Survey B participants randomly into smaller groups (each different price).

Randomly splitting between Survey A and Survey B

- A simple way to split people between the two surveys is to use a random link redirect tool like [Allocate Monster](#). It randomly sends someone to Survey A or Survey B when they click the survey link.
- When your surveys are ready, insert the links of the two surveys in the tool and generate the link you will use for your survey. It will look something like this: <https://allocate.monster/XXXXXX>
- **Note:** Allocate Monster only accepts links to known survey software to avoid misuse.



Feed me survey links!

Enter the two or three survey-based links you want to randomly redirect to in the box below.

Please note that the link created is public, so it is up to you to set up privacy controls for any information you share through the link.

Researchers, please cite the use of this tool with the following:

Fergusson, A. (2016). Designing online experiments using Google forms+ random redirect tool. Retrieved from <https://teaching.statistics-is-awesome.org/designing-online-experiments-using-google-forms-random-redirect-tool>

Note: Only links with [domains on the approved list](#) will be accepted! Please [complete this form](#) to request a domain to be added to the approved list.

Link 1

Link 2

Link 3

Get random redirect link

Sold to

brian@briancliette.me



Survey software

- We will use [Qualtrics](#). Most survey software does not offer question randomization in the way we need it, this is one of the few that does. It's commonly used by researchers.
- **Note:** Qualtrics has a free version, but you will need randomization which is a premium feature. The premium 30 day free trial has a limit of 300 responses, which is not enough.

Survey A - Open question

- Create a new survey and write a description of the product. If possible, use images or video. You can even provide a sample of the product.
- Insert a question asking "How much would you pay - maximum - for [insert the name of your product]?" (or a similar variation). Make sure to specify the currency you are looking for. Allow respondents to insert any amount.

▼

Open question

...

Intro

Introducing: **Ariyh Playbooks**

PDFs of 20 to 40 pages that are:

- **Practical.** Intensive, example-packed guide designed for results. No fluff.
- **Latest science focused.** Apply the latest research, not outdated knowledge.
- **Lifetime updates.** Updated when significant new discoveries are made.

Plus, they come with a:

- **Results guarantee.** Applied the playbook and didn't get results? Get a refund.

Price open

★

How much would you pay - maximum - for a Playbook? (in USD \$)

Survey B - Closed question

- Duplicate Survey A. Everything should be identical except for the price question.
- In the same 'block' of questions (as Qualtrics calls it), insert all the different closed questions with the prices you want to ask about.
- Ask "Would you buy [insert the name of your product] for [one of your chosen prices]?". People should only be able to answer either "Yes" or "No".
- Remember they should be minimum 7 different prices, start and end with the extremes of your possible pricing range, and have equal increments between them.

Price 10



Would you pay \$10 for a Playbook?

- ☐ Yes
☐ No

☐ Price 20



Would you pay \$20 for a Playbook?

- ☐ Yes
☐ No

☐ Price 30

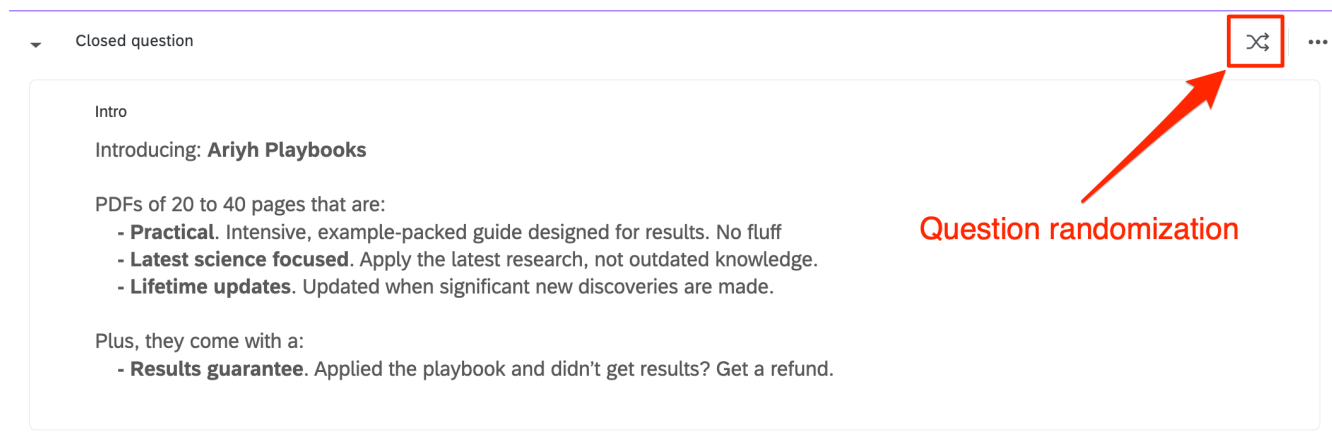


Would you pay \$30 for a Playbook?

- ☐ Yes
☐ No

Randomization of Survey B questions

- Click on the top right part of the questions block (or on the left side of the page, if shown under “Block behavior”).



Closed question

Intro

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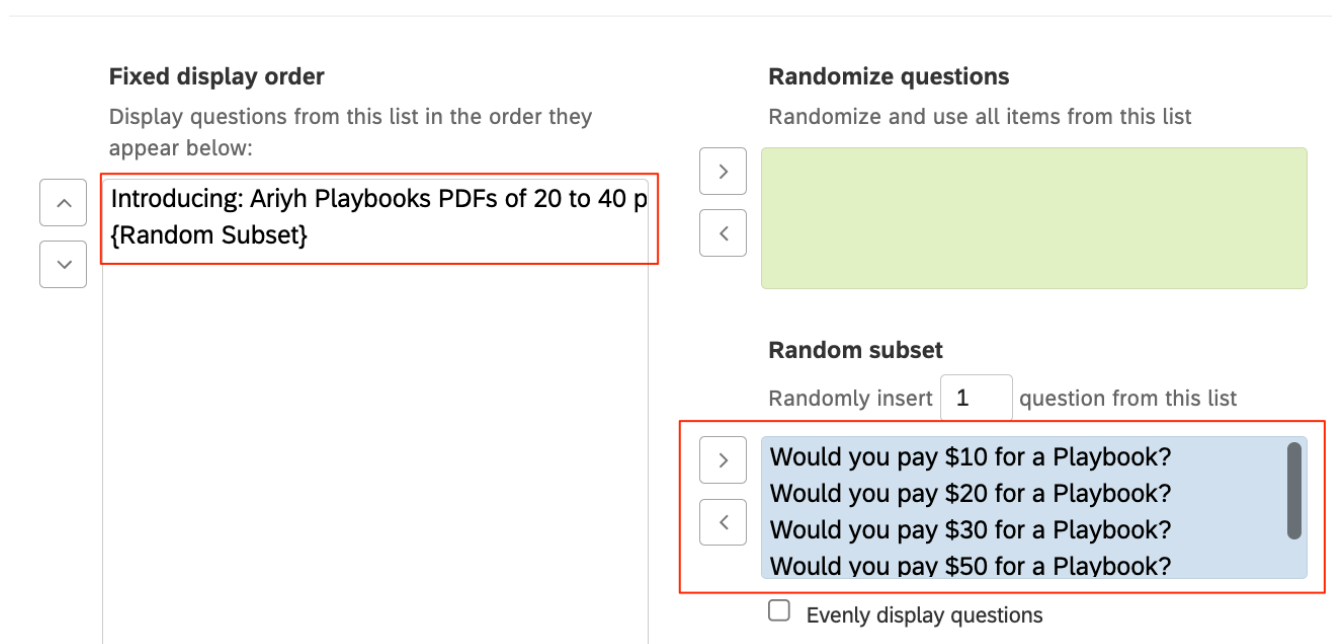
- **Practical.** Intensive, example-packed guide designed for results. No fluff
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- **Lifetime updates.** Updated when significant new discoveries are made.

Plus, they come with a:

- **Results guarantee.** Applied the playbook and didn't get results? Get a refund.

- Choose *Advanced Randomization* and click on *Set Up Advanced Randomization*
- Move all the pricing questions into the *Random subset* box. Set the *Randomly insert [] question from this list to 1*. You don't need to check the *Evenly display questions* box.
- In the *Fixed display order* box on the left make sure the *{Random subset}* question is after the description. Save and exit.

Advanced Randomization



Fixed display order

Display questions from this list in the order they appear below:

- Introducing: Ariyh Playbooks PDFs of 20 to 40 p {Random Subset}

Randomize questions

Randomize and use all items from this list

Random subset

Randomly insert question from this list

- Would you pay \$10 for a Playbook?
- Would you pay \$20 for a Playbook?
- Would you pay \$30 for a Playbook?
- Would you pay \$50 for a Playbook?

☐ Evenly display questions

Testing

- Preview each survey to see if they work as desired. Survey B should randomly show different price questions. If all is working, go ahead and publish them.
- Insert the two links in [Allocate Monster](#) (as we saw above) and generate your unique survey link.
- Try the link in an incognito window of your browser a few times. You should randomly see different survey versions and prices. Fill out a few of them.
- Check that the answers were recorded. Then go ahead and delete them in Qualtrics.
- We are now ready to run the survey.

Run the survey

- Share the survey with a large enough sample that's representative of your target customers. Expect that only a small percentage of people will fill it out. Offer incentives to increase that amount.
- In Ariyh's case, the survey was shared in [Ariyh's newsletter](#) and 3 random respondents would win a free playbook.

Analyzing the results

Prepare the results

- When you have enough responses, close the surveys and export results from Qualtrics to a spreadsheet.
- Remove outliers such as \$0 and unrealistic numbers (e.g. \$236 for a cookie, \$10 for a car) from Survey A results.
- **Note:** Ariyh's survey had 276 responses, which was too low and makes results less accurate. It would have been better to use fewer price options in Survey B (7 instead of 10) and keep it open for longer (2 or 3 weeks). Learn from my mistakes

Combine results from Survey A and Survey B

- Calculate the percentage of yes for each question in Survey B ("Would you buy [product] for \$X? [Yes/No]"). Then, use this proportion, out of the total of respondents, to calculate the average price from this survey.

Closed question results

Price	\$10	\$20	\$30	\$40	\$50	\$60	\$70	\$80	\$90	\$100
Percent yes	73%	87%	82%	63%	43%	43%	47%	40%	43%	47%
Yes	11	13	14	10	6	6	7	6	6	7
No	4	2	3	6	8	8	8	9	8	8
Total	15	15	17	16	14	14	15	15	14	15
Avg	\$7.33	\$17.33	\$24.71	\$25.00	\$21.43	\$25.71	\$32.67	\$32.00	\$38.57	\$46.67
Average price	\$27.14									

- Calculate the average price from all respondents in Survey A (“How much would you pay at a maximum?”).
- Then, for each price given by respondents in Survey A:
 - Subtract the average price of Survey A
 - Add the average price of Survey B
 - For example, an answer is \$50, subtract the Survey A average of \$28.85 and add the Survey B average of \$27.14. The result is \$48.3.
- You now have the true price each respondent in Survey A would pay.

Plotting your expected sales at different prices

- Choose a few different price options and look at what percentage of people would buy each. For example, 61.9% would buy it if it's priced at \$18, but only 28.6% would if it's priced at \$35.
- Calculate how much total profit that would generate per sale (remember to subtract your per unit cost if necessary).
- Congratulations, you can now easily choose the optimal price for your product!

Willingness to pay at different price points

Price	\$5	\$8	\$12	\$18	\$25	\$35	\$45	\$55	\$73	\$89
Yes	116	114	88	78	42	36	28	14	12	8
Percent yes	92.06%	90.48%	69.84%	61.90%	33.33%	28.57%	22.22%	11.11%	9.52%	6.35%
Revenue sample	\$580	\$912	\$1,056	\$1,404	\$1,050	\$1,260	\$1,260	\$770	\$876	\$712

Choosing your price

- Say that you find that:
 - 80% of people are willing to buy your product for \$50
 - 50% are willing to buy it for \$80
 - 20% are willing to buy it for \$140
- If your audience is 10,000 people, and your product costs \$20 to make, that would mean:
 - 8,000 sales x \$30 (\$50 - \$20) = \$240,000 profit
 - 5,000 sales x \$60 (\$80 - \$20) = \$300,000 profit
 - 2,000 sales x \$120 (\$140 - \$20) = \$240,000 profit
- So \$80 would be the best price to choose. But this shouldn't always be your choice.
- Think of the long-term signals you send. For example, if you are trying to cultivate a luxury brand image, you may be better off choosing a price of \$140. Even if immediate profits are lower, you will attract customers that are more likely to buy future products.
- In the case of Ariyh, I chose a price on the lower end of the spectrum to generate a larger number of sales (but not too low that it would seem low quality). This way, more people get to know about Ariyh (including [the newsletter](#)) and can apply scientific research to grow their business.



Thank you

Thank you for reading and purchasing Ariyh's Science-based Pricing & Promotions Playbook.

Did you enjoy it or have suggestions? I'd love to hear from you.

Email me at thomas@ariyh.com

And please let others know too by leaving your review [on Gumroad](#).

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